

Inter-Port Crosstalk Reduction in Single Multi-Input-Port WSS-Based ROADM Architectures Using Interleaved Port Allocation

Masahiko Jinno ^{(1),(2)}, Ryunosuke Sasaki ⁽²⁾, Emre Maleeq Bin Mohd Rashdan ⁽²⁾, Takashi Miyamura ⁽¹⁾, Satoru Okamoto ⁽¹⁾, and Naoaki Yamanaka ⁽¹⁾

⁽¹⁾ Keio Frontier Research & Education Collaborative Square at Shin-Kawasaki, Keio University, jinno.masahiko@kagawa-u.ac.jp

⁽²⁾ Faculty of Engineering and Design, Kagawa University

Abstract We analyse joint-switching-induced inter-port crosstalk (XT) in single multi-input-port WSS-based ROADM architectures. We propose an interleaved port allocation scheme that suppresses XT without additional hardware and validate it experimentally, demonstrating considerable XT reduction while preserving the simplicity and cost advantages of the single-WSS architecture. ©2026 The Authors

Introduction

To address the increasing capacity demands and cost constraints in metro optical networks, hardware-efficient optical node architectures with reduced complexity and power consumption have been actively explored [1]. Among these, reconfigurable optical add/drop multiplexers (ROADMs) leveraging multi-input-port wavelength-selective switches (MIP-WSSs) have been proposed to simplify node architectures [2,3]. Building on this concept, we have previously demonstrated a colourless and directionless (CD) ROADM configuration based on a single MIP-WSS [4]. By integrating all ROADM functionalities into a single switching element, the proposed architecture significantly reduces equipment cost, power consumption, footprint, intra-node fibre connections, and insertion loss, while maintaining the essential flexibility of WSS-based CD-ROADMs.

In addition, the wavelength contention inherent in MIP-WSS operation can be alleviated by joint switching, which was initially introduced in the context of space-division multiplexing (SDM) WSSs, where spatially multiplexed subchannels (spatial superchannel) are jointly switched [5]. In this technique, a single wavelength-selective switching mirror is shared to simultaneously route multiple optical signals at the same wavelength originating from different input ports. As a result, even in a single MIP-WSS-based ROADM, the same wavelength can be assigned to jointly switched optical signals, including both upstream and downstream signals transmitted between any ROADM port pair [4]. While this approach improves wavelength utilisation and relaxes contention constraints [6], it inherently introduces inter-port in-band crosstalk (XT) among jointly switched optical signals. This in-band XT may become a critical performance bottleneck yet has not been sufficiently addressed in prior work.

To the best of our knowledge, this is the first work to analyse and mitigate joint-switching-induced inter-port XT in single MIP-WSS-based

ROADMs. We propose an interleaved port allocation scheme that suppresses XT without additional hardware and validate it experimentally.

Architecture and XT Mechanism of Conventional CD-ROADM

In a conventional CD-ROADM, each direction employs a single-input-port (SIP) WSS, and the add/drop section consists of an SIP-WSS and a splitter/coupler (SPL/CPL). Because the SIP-WSS and SPL/CPL are connected by a single fibre, light paths traversing this section must use different wavelengths, resulting in a wavelength contention constraint in the add/drop section.

Inter-port XT arising from leakage in WSSs occurs only between signals at the same wavelength that traverse different paths [7]. No XT occurs between upstream and downstream signals assigned the same wavelength and traversing the same path, as they are routed through different WSSs. Furthermore, in route-and-select (R&S) architectures, this XT is effectively suppressed by cascaded blocking at both ingress and egress WSSs, making it negligible and largely insignificant in conventional R&S ROADM operation, albeit at the expense of spectral narrowing.

Architecture and XT Mechanism of Single MIP-WSS-Based CD-ROADM

Figure 1(a) shows the single MIP-WSS-based degree-3 CD-ROADM reported in [4], which is implemented by reusing a standard 1×9 WSS with ten equally spaced single-mode fibres. The figure simplifies the WSS structure by omitting the wavelength-demultiplexing section, which typically comprises a diffraction grating and associated optics, leaving only the beam-steering section for clarity. A set of flat mirrors is arranged along the wavelength axis—one for each wavelength channel.

The first four fibres (F_0 – F_3) serve as inputs and the remaining four (F_4 – F_7) as outputs, forming a 4×4 WSS, as illustrated in Fig. 1(b). This

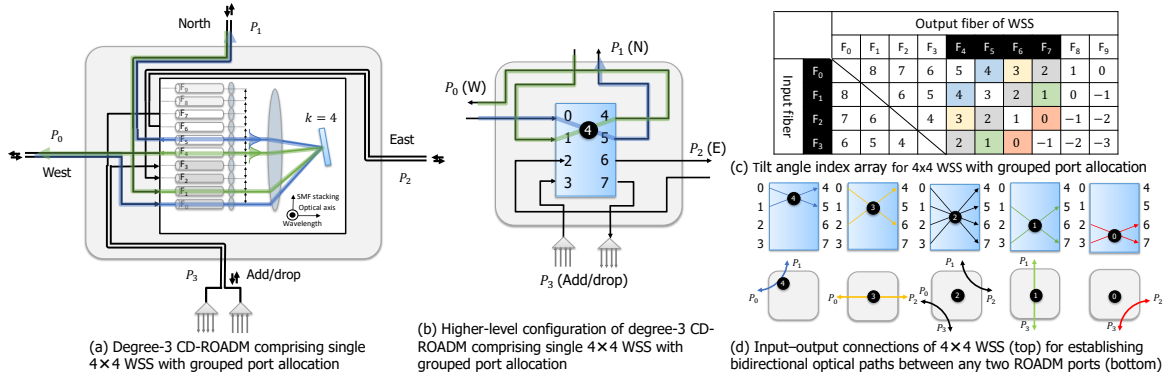


Fig. 1: Architecture and XT mechanism of a single MIP-WSS-based CD-ROADM with grouped port allocation.

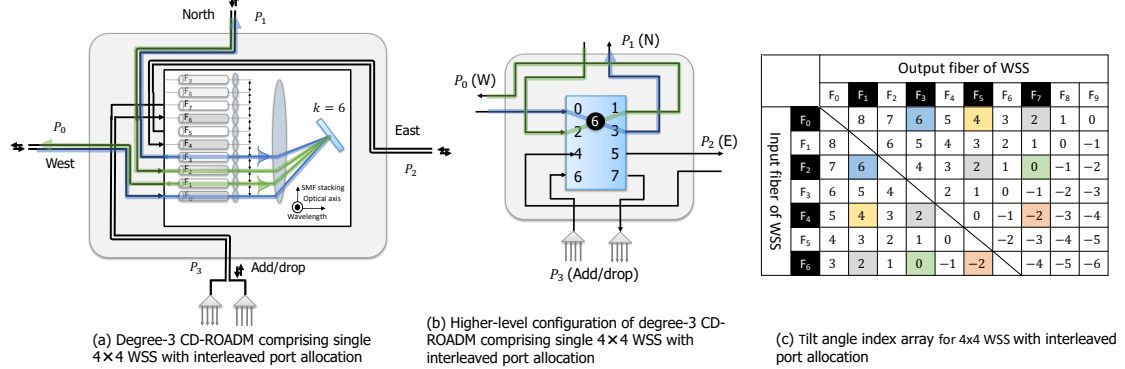


Fig. 2: Architecture and XT mechanism of a single MIP-WSS-based CD-ROADM with interleaved port allocation.

simple port allocation scheme is hereafter referred to as *grouped port allocation*. The three line-side ports (P_0 – P_2) and one add/drop port (P_3) of the ROADM are connected to the eight fibres of the WSS according to the following rule: when the i -th fibre is connected to the input side of a port, the output side of that port is connected to the $(i + 4)$ -th fibre. The add/drop port P_3 is connected to an SPL/CPL pair that aggregates and distributes signals to client devices. In addition to wavelength contention within the MIP-WSS, contention also occurs on the fibre between the SPL/CPL and the MIP-WSS, corresponding to the same add/drop contention as in conventional CD-ROADMs.

Figure 1(c) shows the array $K_{4 \times 4}$, where each element k_{ij} represents the normalized tilt-angle index of the WSS required to connect F_i and F_j , and is given by $k_{ij} = N - 1 - (i + j)$, where N ($=8$) is the total number of used fibres in the WSS. Identical values appear along each diagonal of the array, indicating that the corresponding connections share the same mirror tilt angle. Thus, a single wavelength-specific mirror can steer multiple same-wavelength beams from different input fibres. This operation is referred to as joint switching, and same-coloured cells in the array indicate joint-switchable connections.

The correspondence rule between the ROADM ports and WSS fibres described above is designed to enable the same wavelength to be assigned to upstream and downstream optical

signals established between any pair of ports by alleviating wavelength contention in the MIP-WSS through joint switching. Figure 1(d) illustrates the connections between the input and output fibres of the 4×4 WSS (top) required to establish bidirectional optical paths between any two ports of the ROADM (bottom). The numbers shown in the black circles indicate the mirror tilt angles used for the corresponding joint switching in the WSS.

The *grouped port allocation* is straightforward; however, the adjacent arrangement of output fibre ports makes it susceptible to inter-port XT, as illustrated by the Gaussian beam profiles of upstream and downstream signals along the west (P_0) to north (P_1) path in Fig. 1(a). To address this issue without increasing hardware complexity, we propose *interleaved port allocation*. This scheme was originally introduced in SDM WSSs for spatial super-channels, as a means to reduce mirror tilt angles [8,9]. In the single MIP-WSS-based ROADM, even-numbered fibres (F_0, F_2, F_4, F_6) of the MIP-WSS serve as inputs and odd-numbered fibres (F_1, F_3, F_5, F_7) as outputs, as shown in Fig. 2, thereby increasing output fibre port spacing and mitigating inter-port XT by reducing the spatial overlap between adjacent Gaussian beams (Fig. 2(a)). As shown in the mirror tilt angle index array of Fig. 2(c), interleaved port allocation preserves joint-switching capability between the required fibre ports.

Inter-Port XT Measurements for Grouped and Interleaved Port Allocations

We measured inter-port XT between jointly switched optical signals in a CD-ROADM comprising a single 4×4 WSS configured from a commercial 1×9 WSS with a liquid-crystal-on-silicon (LCoS) spatial light modulator. Measurements were conducted for both grouped and interleaved port allocations across eight connection patterns between the input and output fibres of the 4×4 WSS, corresponding to bidirectional optical paths between arbitrary ROADM port pairs.

As shown in Fig. 3, grouped port allocation leads to relatively high XT for most adjacent output port pairs (five cases on the left), with a worst-case value of -32 dB. Here, $XT(i, j, k)$ denotes the XT from fibre i to fibre k when routing from i to j , defined in decibels as the ratio of the leaked optical power at fibre k to the output power at fibre j . With interleaved port allocation, the XT is reduced to approximately -39 dB or lower in most cases. The absence of data corresponding to $XT(3, 6, 7)$ and $XT(3, 7, 6)$ is due to the lack of support for negative tilt angles in the used WSS. In contrast, for non-adjacent output port pairs, the XT is already low even with grouped port allocation, and no significant improvement is observed with interleaving. A few exceptions are observed, likely attributable to higher-order diffraction arising from pixelation and phase discontinuities in the sawtooth phase profiles of the WSS. While interleaved port allocation effectively reduces inter-port XT, the results suggest that further refinement may be required for practical deployment.

Experimental Validation via OSNR Penalty Measurements of 200-G DP-16QAM signals

When in-band XT exceeds a certain threshold, the required optical signal-to-noise ratio (OSNR) to achieve a given bit-error ratio (BER) increases rapidly. This threshold decreases as the order of quadrature amplitude modulation (QAM) increases [10]. For example, a digital coherent transceiver used in our experiments exhibits a 0.5-dB OSNR penalty at a BER of 1×10^{-3} for in-band XT levels of -22 dB and -29 dB for 31.6-Gbaud dual-polarisation (DP) QPSK (100 Gb/s) and 16QAM (200 Gb/s), respectively.

Since the worst-case inter-port XT measured for the single MIP-WSS-based ROADM is -32 dB even with grouped port allocation, the resulting OSNR penalty for optical signals generated by this digital coherent transceiver remains negligibly small after a single pass through the ROADM. However, inter-port XT accumulates over successive ROADM traversals and can become non-negligible in multi-hop scenarios. A 7 dB reduction in inter-port XT (from -32 dB to -39 dB) enables approximately a fivefold increase in the number of ROADM nodes that can be traversed.

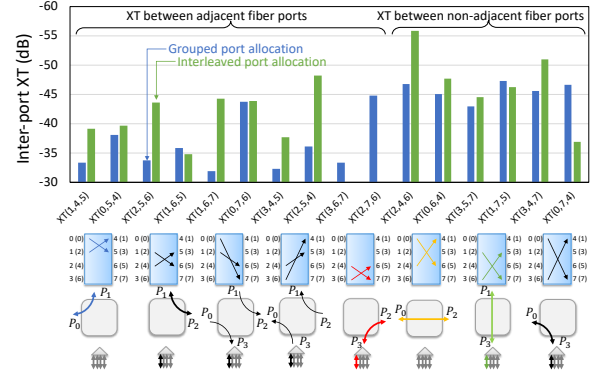


Fig. 3: Measured inter-port XT for jointly switched optical signals in grouped and interleaved port allocations.

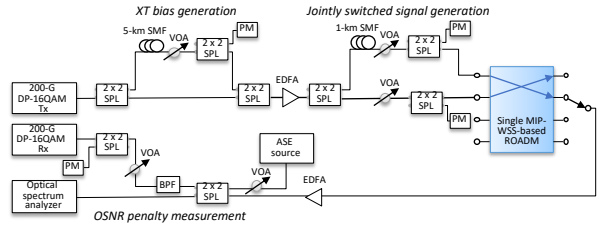


Fig. 4: Experimental setup for OSNR penalty measurements.

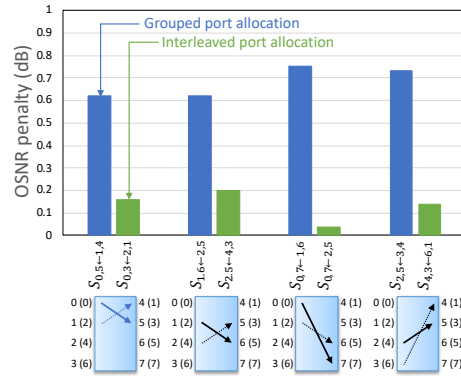


Fig. 5: OSNR penalties for jointly switched signals.

Using the setup shown in Fig. 4, we experimentally investigated the OSNR penalty improvement by emulating multiple ROADM traversals using 200-Gb/s DP-16QAM optical signals with a predefined in-band XT of -30 dB. The OSNR penalty for adjacent jointly switched output ports is compared in Fig. 5, confirming the effectiveness of the proposed interleaved port allocation scheme. Here, $S_{i,j \leftarrow k,l}$ denotes the optical signal routed from port i to j , jointly switched with the signal from port k to l and subject to its XT.

Conclusions

We analysed joint-switching-induced inter-port XT in single MIP-WSS-based ROADM architectures. We proposed an interleaved port allocation scheme that suppresses XT without increasing hardware complexity and validated its effectiveness through OSNR penalty measurements using 200-Gb/s DP-16QAM optical signals, demonstrating considerable inter-port XT reduction in most cases while preserving the simplicity and cost advantages of the single-WSS architecture.

Acknowledgements

The authors thank santec AOC Corporation for the use of their LCoS-based WSS equipment. This work was supported in part by the NICT (JPJ012368C08501).

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